Course Code	EC24P
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Class	T.E – EXTC. SEM-V
Division	B
Date of Performance	12th September 2025
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EXPERIMENT NO.7

Estimated time to complete this experiment: 02 hours.

Objective: To simulate and find the propagation delay time of symmetric CMOS inverter using gpdk180.

CO to be achieved: CO2.

Expected Outcome of Experiment:

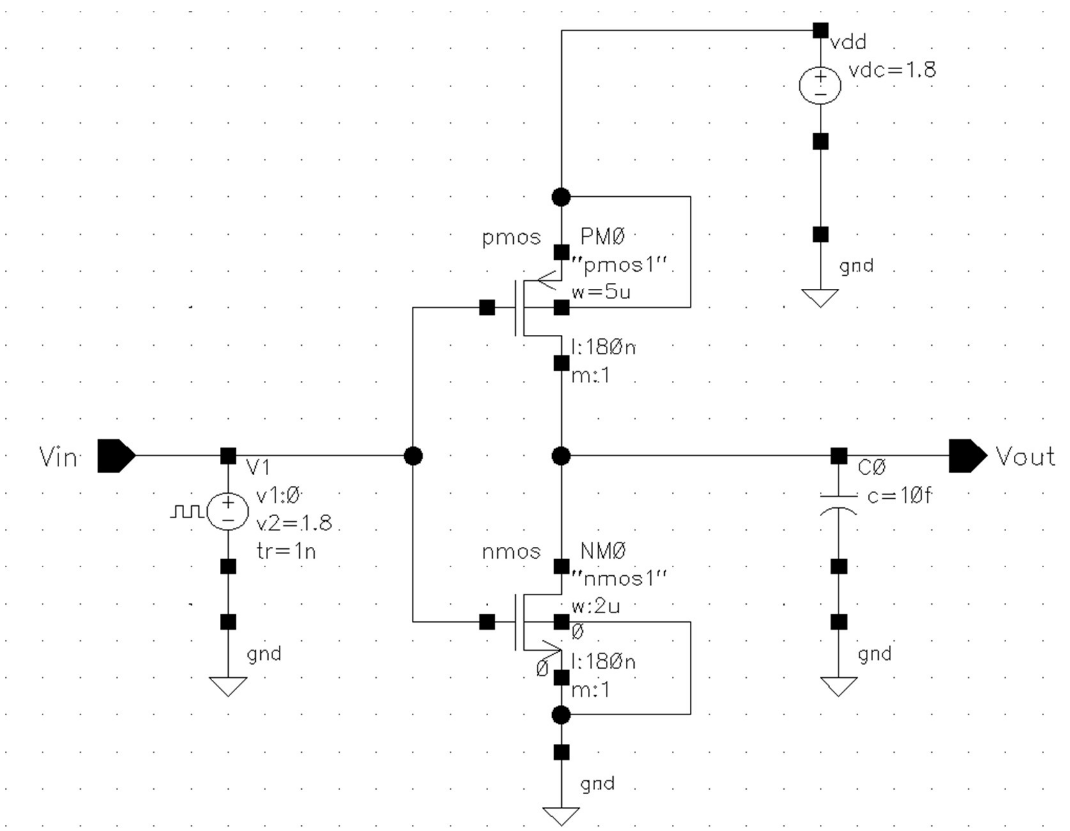
- Students will learn cadence virtuoso.
- Students will be able to find rise time and fall time of asymmetric CMOS inverter.
- Students will be able to find propagation delay time of CMOS Invereter.

Pre Lab/ Prior Concept: Knowledge of working of MOSFET and CMOS inverter transient analysis.

Apparatus/TOOL:

- PC
- CADENCE VIRTUOSO.

Circuit Diagram:



Procedure.

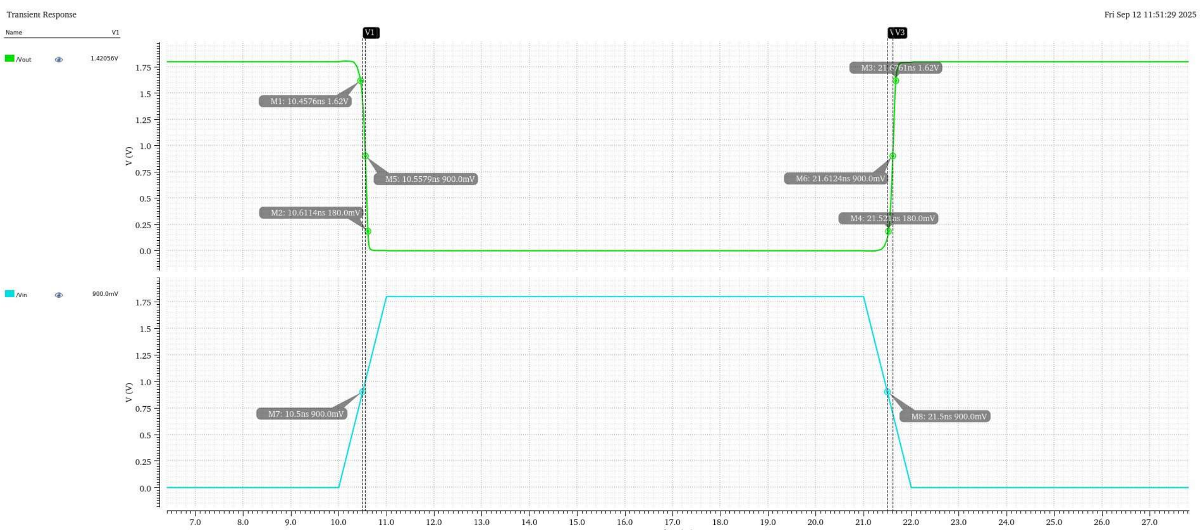
- Open the cadence virtuoso tool in Red-Hat Linux. The commands are
  - csh
  - source /home/install/cshrc
  - virtuoso.
- Create the Library and cell view
- Open log window using terminal.
- Click on the new option, now create library with gpdk180 from technology library.
- Again click the on new option, and set the cell view.
- Cadence Virtuoso dashboard will be opened.

<u>For characterization of 180nm CMOS Inverter.</u>

- Select NMOS transistor from add instance and select its channel length as 180nm (Nano meter) and channel width W=2 um (micro-meter).
- Select PMOS transistor from add instance and select its channel length as 180nm (Nano meter) and channel width as W= 5um (micro-meter).
- Construct the circuit shown in circuit diagram in Cadence.

- Take VDD=1.8V.
- Apply input as V_{PULSE} with following specification.
  - *Voltage 1=0V*
  - *Voltage 2=1.8V*
  - *Delay time=10ns*
  - *Rise time =1ns*
  - *Fall time =1ns*
  - *Pulse width = 10ns.*
  - *Period = 20ns.*
- Then perform the Transient analysis and plot the input and output waveform.
- Using input and output waveform calculate the following parameters.
  - T_{RISE}, T_{FALL}, T_{PHL}, T_{PLH}
  - Propagation delay time $T_D = \frac{TPHL+TPLH}{2}$

Results:



Observation.

- T_{RISE} = *0.1538*
- T_{FALL} = *0.1551*
- T_{PHL} = *0.0579*
- T_{PLH} = *0.1124*
- Propagation delay time $T_D = \frac{TPHL+TPLH}{2} = \frac{(0.0579+\ .1124)}{2} = 0.0851$

Conclusion:

The measured rise time (0.1538 ns) and fall time (0.1551 ns) are nearly identical, showing that the CMOS inverter achieves balanced switching for both transitions. The propagation delay values (TPHL = 0.0579 ns and TPLH = 0.1124 ns) indicate that the inverter switches faster from high to low than from low to high. The average propagation delay of 0.0851 ns highlights the inverter's overall efficiency and confirms reliable performance in 180 nm technology. These results demonstrate that the inverter delivers fast and stable switching, making it well-suited for high-speed digital applications.

Post-Lab Question

Q1) What are the practical applications of a symmetric CMOS inverter?

A symmetric CMOS inverter, designed with well-balanced PMOS and NMOS transistors, is preferred for its stability and efficiency. Some of its major applications are:

- Digital Logic Design: Serves as the foundation for gates like NAND, NOR, and XOR in CPUs and memory systems.
- Signal Integrity: Provides quick, clean, and consistent switching.
- Clock Circuits: Used for generating and buffering accurate clock signals.
- Voltage Level Conversion: Facilitates level shifting between different circuit sections.
- Oscillators and Timing Systems: Plays a role in ring oscillators and other timing circuits.
- Low-Power Electronics: Suitable for portable and battery-operated devices due to its energy efficiency

Practical Performance (5 Marks)	Presentation and Conclusion (5 Marks)	Total (10 Marks)	Sign	Date of Submission.